

The heat, Q , required to change the phase of a sample of mass m is

$$Q = mL_f \text{ (for melting/freezing),}$$

$$Q = mL_v \text{ (for vaporization/condensation),}$$

where L_f is the **latent heat of fusion**, and L_v is the **latent heat of vaporization**. The latent heat of fusion is the amount of heat needed to cause a phase change between solid and liquid. The latent heat of vaporization is the amount of heat needed to cause a phase change between liquid and gas. L_f and L_v are coefficients that vary from substance to substance, depending on the strength of intermolecular forces, and both have standard units of J/kg. See [Table 11.3](#) for values of L_f and L_v of different substances.

Substance	Melting Point (°C)	L_f (kJ/kg)	Boiling Point (°C)	L_v (kJ/kg)
Helium	−269.7	5.23	−268.9	20.9
Hydrogen	−259.3	58.6	−252.9	452
Nitrogen	−210.0	25.5	−195.8	201
Oxygen	−218.8	13.8	−183.0	213
Ethanol	−114	104	78.3	854
Ammonia	−78	332	−33.4	1370
Mercury	−38.9	11.8	357	272
Water	0.00	334	100.0	2256
Sulfur	119	38.1	444.6	326
Lead	327	24.5	1750	871
Antimony	631	165	1440	561
Aluminum	660	380	2520	11400
Silver	961	88.3	2193	2336
Gold	1063	64.5	2660	1578
Copper	1083	134	2595	5069
Uranium	1133	84	3900	1900
Tungsten	3410	184	5900	4810

TABLE 11.3 Latent Heats of Fusion and Vaporization, along with Melting and Boiling Points

Let's consider the example of adding heat to ice to examine its transitions through all three phases—solid to liquid to gas. A phase diagram indicating the temperature changes of water as energy is added is shown in [Figure 11.10](#). The ice starts out at $-20\text{ }^{\circ}\text{C}$, and its temperature rises linearly, absorbing heat at a constant rate until it reaches $0\text{ }^{\circ}\text{C}$. Once at this temperature, the ice gradually melts, absorbing 334 kJ/kg . The temperature remains constant at $0\text{ }^{\circ}\text{C}$ during this phase change. Once all the ice has melted, the temperature of the liquid water rises, absorbing heat at a

new constant rate. At 100 °C, the water begins to boil and the temperature again remains constant while the water absorbs 2256 kJ/kg during this phase change. When all the liquid has become steam, the temperature rises again at a constant rate.

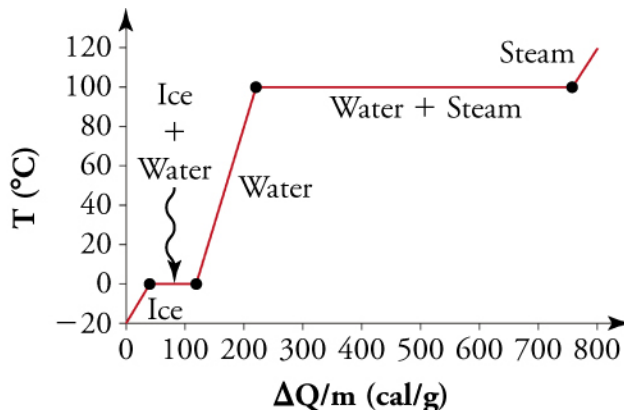


FIGURE 11.10 A graph of temperature versus added energy. The system is constructed so that no vapor forms while ice warms to become liquid water, and so when vaporization occurs, the vapor remains in the system. The long stretches of constant temperature values at 0 °C and 100 °C reflect the large latent heats of melting and vaporization, respectively.

We have seen that vaporization requires heat transfer to a substance from its surroundings. Condensation is the reverse process, where heat is transferred *away from* a substance *to* its surroundings. This release of latent heat increases the temperature of the surroundings. Energy must be removed from the condensing particles to make a vapor condense. This is why condensation occurs on cold surfaces: the heat transfers energy away from the warm vapor to the cold surface. The energy is exactly the same as that required to cause the phase change in the other direction, from liquid to vapor, and so it can be calculated from $Q = mL_v$. Latent heat is also released into the environment when a liquid freezes, and can be calculated from $Q = mL_f$.



FUN IN PHYSICS

Making Ice Cream



FIGURE 11.11 With the proper ingredients, some ice and a couple of plastic bags, you could make your own ice cream in five minutes. (ElinorD, Wikimedia Commons)

Ice cream is certainly easy enough to buy at the supermarket, but for the hardcore ice cream enthusiast, that may not be satisfying enough. Going through the process of making your own ice cream lets you invent your own flavors and marvel at the physics firsthand ([Figure 11.11](#)).

The first step to making homemade ice cream is to mix heavy cream, whole milk, sugar, and your flavor of choice; it could be as simple as cocoa powder or vanilla extract, or as fancy as pomegranates or pistachios.

The next step is to pour the mixture into a container that is deep enough that you will be able to churn the mixture without it spilling over, and that is also freezer-safe. After placing it in the freezer, the ice cream has to be stirred vigorously every 45 minutes for four to five hours. This slows the freezing process and prevents the ice cream from turning into a solid block of ice. Most people prefer a soft creamy texture instead of one giant popsicle.

As it freezes, the cream undergoes a phase change from liquid to solid. By now, we're experienced enough to know that this means that the cream must experience a loss of heat. Where does that heat go? Due to the temperature difference between the freezer and the ice cream mixture, heat transfers thermal energy from the ice cream to the air in the freezer. Once the temperature in the freezer rises enough, the freezer is cooled by pumping excess heat outside into the kitchen.

A faster way to make ice cream is to chill it by placing the mixture in a plastic bag, surrounded by another plastic bag half full of ice. (You can also add a teaspoon of salt to the outer bag to lower the temperature of the ice/salt mixture.) Shaking the bag for five minutes churns the ice cream while cooling it evenly. In this case, the heat transfers energy out of the ice cream mixture and into the ice during the phase change.

This [video \(http://www.openstax.org/l/28icecream\)](http://www.openstax.org/l/28icecream) gives a demonstration of how to make home-made ice cream using ice and plastic bags.

Why does the ice bag method work so much faster than the freezer method for making ice cream?

- Ice has a smaller specific heat than the surrounding air in a freezer. Hence, it absorbs more energy from the ice-cream mixture.
- Ice has a smaller specific heat than the surrounding air in a freezer. Hence, it absorbs less energy from the ice-cream mixture.
- Ice has a greater specific heat than the surrounding air in a freezer. Hence, it absorbs more energy from the ice-cream mixture.
- Ice has a greater specific heat than the surrounding air in a freezer. Hence, it absorbs less energy from the ice-cream mixture.

Solving Thermal Energy Problems with Phase Changes



WORKED EXAMPLE

Calculating Heat Required for a Phase Change

Calculate a) how much energy is needed to melt 1.000 kg of ice at 0 °C (freezing point), and b) how much energy is required to vaporize 1.000 kg of water at 100 °C (boiling point).

Strategy FOR (A)

Using the equation for the heat required for melting, and the value of the latent heat of fusion of water from the previous table, we can solve for part (a).

Solution to (a)

The energy to melt 1.000 kg of ice is

$$Q = mL_f = (1.000 \text{ kg}) (334 \text{ kJ/kg}) = 334 \text{ kJ.} \quad 11.18$$

Strategy FOR (B)

To solve part (b), we use the equation for heat required for vaporization, along with the latent heat of vaporization of water from the previous table.

Solution to (b)

The energy to vaporize 1.000 kg of liquid water is

$$Q = mL_v = (1.000 \text{ kg})(2256 \text{ kJ/kg}) = 2256 \text{ kJ.} \quad 11.19$$

Discussion

The amount of energy need to melt a kilogram of ice (334 kJ) is the same amount of energy needed to raise the temperature of 1.000 kg of liquid water from 0 °C to 79.8 °C . This example shows that the energy for a phase change is enormous compared to energy associated with temperature changes. It also demonstrates that the amount of energy needed for vaporization is even greater.



WORKED EXAMPLE

Calculating Final Temperature from Phase Change: Cooling Soda with Ice Cubes

Ice cubes are used to chill a soda at 20 °C and with a mass of $m_{\text{soda}} = 0.25 \text{ kg}$. The ice is at 0 °C and the total mass of the ice cubes is 0.018 kg. Assume that the soda is kept in a foam container so that heat loss can be ignored, and that the soda has the same specific heat as water. Find the final temperature when all of the ice has melted.

Strategy

The ice cubes are at the melting temperature of 0 °C . Heat is transferred from the soda to the ice for melting. Melting of ice occurs in two steps: first, the phase change occurs and solid (ice) transforms into liquid water at the melting temperature; then, the temperature of this water rises. Melting yields water at 0 °C, so more heat is transferred from the soda to this water until they are the same temperature. Since the amount of heat leaving the soda is the same as the amount of heat transferred to the ice.

$$Q_{\text{ice}} = -Q_{\text{soda}} \quad 11.20$$

The heat transferred to the ice goes partly toward the phase change (melting), and partly toward raising the temperature after melting. Recall from the last section that the relationship between heat and temperature change is $Q = mc\Delta T$. For the ice, the temperature change is $T_f - 0 \text{ °C}$. The total heat transferred to the ice is therefore

$$Q_{\text{ice}} = m_{\text{ice}}L_f + m_{\text{ice}}c_w(T_f - 0 \text{ °C}). \quad 11.21$$

Since the soda doesn't change phase, but only temperature, the heat given off by the soda is

$$Q_{\text{soda}} = m_{\text{soda}}c_w(T_f - 20 \text{ °C}). \quad 11.22$$

Since $Q_{\text{ice}} = -Q_{\text{soda}}$,

$$m_{\text{ice}}L_f + m_{\text{ice}}c_w(T_f - 0 \text{ °C}) = -m_{\text{soda}}c_w(T_f - 20 \text{ °C}). \quad 11.23$$

Bringing all terms involving T_f to the left-hand-side of the equation, and all other terms to the right-hand-side, we can solve for T_f .

$$T_f = \frac{m_{\text{soda}}c_w(20 \text{ °C}) - m_{\text{ice}}L_f}{(m_{\text{soda}} + m_{\text{ice}})c_w} \quad 11.24$$

Substituting the known quantities

$$T_f = \frac{(0.25 \text{ kg})(4186 \text{ J/kg} \cdot \text{°C})(20 \text{ °C}) - (0.018 \text{ kg})(334,000 \text{ J/kg})}{(0.25 \text{ kg} + 0.018 \text{ kg})(4186 \text{ J/kg} \cdot \text{°C})} = 13 \text{ °C} \quad 11.25$$

Discussion

This example shows the enormous energies involved during a phase change. The mass of the ice is about 7 percent the mass of the soda, yet it causes a noticeable change in the soda's temperature.

TIPS FOR SUCCESS

If the ice were not already at the freezing point, we would also have to factor in how much energy would go into raising its temperature up to 0 °C, before the phase change occurs. This would be a realistic scenario, because the temperature of ice is often below 0 °C .